Forestry: An International Journal of Forest Research, 2026, **99**, 1–14
https://doi.org/10.1093/forestry/cpaf053
Advance access publication date 11 September 2025
Original Article

Early detection of bark beetle infestations in Central Europe using deep learning–based reconstructions of irregular Sentinel-2 time series

Christopher Schiller [1,*], Johannes May[2], Randolf Klinke[3], Fabian Ewald Fassnacht[1]

[1]Remote Sensing and Geoinformatics, Institute of Geographical Sciences, Freie Universität Berlin, Malteserstrasse 74-100, 12249 Berlin, Germany
[2]State Enterprise for Forestry and Timber North Rhine-Westphalia, Albrecht-Thaer-Straße 34, 48147 Münster, Germany
[3]LUP - Luftbild Umwelt Planung GmbH, Große Weinmeisterstraße 3a, 14469 Potsdam, Germany

*Corresponding author. Remote Sensing and Geoinformatics, Institute of Geographical Sciences, Freie Universität Berlin, Malteserstrasse 74-100, 12249 Berlin, Germany. E-mail: christopher.schiller@fu-berlin.de

Abstract

Norway spruce (*Picea abies*) is among the most abundant tree species in Central Europe. Due to climate change-induced extreme weather events, spruce trees are increasingly stressed and therefore threatened by European spruce bark beetle (*Ips typographus*) infestations. Recent mass outbreaks led to severe ecological and economic damage in Central European forests. After an infestation, the filial generation of the beetles swarms out within 6 to 10 weeks to infest new trees. Consequently, an efficient bark beetle management needs to remove infested trees within 10 weeks to prevent further dispersal. While remote sensing allows for large-scale monitoring of forests, the detection of bark beetle infestations remains challenging, as many trees show no visible signs of the infestation within the 10-week detection period. Here, we try to achieve early detections by adjusting a state-of-the-art Deep Learning model to be able to cope with irregular Sentinel-2 satellite time series for reconstruction-based anomaly detection. The model is trained on >300 000 time series of undisturbed coniferous forest and the threshold denoting an anomaly is derived independently, i.e. not from the test dataset. We test the model on a geographically independent dataset with known infestation dates. It achieves moderate performance for detections within 10 weeks after the infestation with a producer's accuracy (PA) of 11.8% ± 8.4% and user's accuracy (UA) of 43.5% ± 24.5% across three model runs, but yields very good results when extending the detection period to 13 weeks (UA = 84.5% ± 7.6%, PA = 81.5% ± 1%). Since the model responds immediately to an anomaly, we conclude that area-wide bark beetle detections within 10 weeks after infestation are likely impossible using Sentinel-2 alone. Still, our approach can readily be used as a near real-time monitoring system for coniferous forest, be applied on any forest disturbance detection task, and may complement terrestrial surveys in the future.

Keywords: remote sensing; anomaly detection; deep learning; bark beetle; disturbance; early detection

Introduction

One of the most abundant tree species in Central European forests is Norway spruce (*Picea abies* L.) (Caudullo et al. 2016), which is often planted in monocultures for timber production. Norway spruce is regularly subject to insect infestations, particularly by the European spruce bark beetle (*Ips typographus* L., in the following: "bark beetle"). Bark beetles benefit from warm and dry weather conditions, allowing them to complete their life cycles more rapidly. As the frequency and intensity of extreme weather events is increasing due to climate change (García-Herrera et al. 2019, Grillakis 2019, Ionita and Nagavciuc 2021), trees are increasingly stressed and their defense mechanisms are weakened. This further promotes bark beetle outbreaks, which are currently becoming more frequent and severe (Senf et al. 2017, Netherer et al. 2019, Patacca et al. 2022). Severe bark beetle infestations have led to mass dieback events (e.g. Harz National Park and Saxon Switzerland National Park in Germany) with significant ecological and economic consequences (Schiefer et al. 2024), which also induced a shift in the species composition in some parts of Central Europe. For instance, the area stocked with spruce in Germany has decreased by ∼17% between 2012 and 2022, much of which is attributed to bark beetle outbreaks (Federal Ministry of Food and Agriculture (BMEL) 2024).

During the first few months after an initial bark beetle infestation, the crowns of spruce trees usually remain green (*green attack* stage) (Ortiz et al. 2013, Kautz 2014). In the green attack stage, the bark beetles establish maternal and brood galleries in the phloem of the tree. The boring beetles also enable wood-degrading fungi to penetrate into the tree (Künniger et al. 2024), which then colonize the sapwood within a few weeks (Solheim 1992). The fungal growth disrupts the water transport through the xylem of the tree (Kirisits and Offenthaler 2002), impairing the tree's defense mechanism of resin production and water transport capabilities. Meanwhile, the beetles lay eggs, which evolve as larvae, pupae and finally beetles, continuously feeding on the phloem of the trees. This leads to a disruption of the nutrient transport in the phloem and consecutively to further breakdown of the tree's defense

Handling editor: Prof. Teja Kattenborn
Received 12 March 2025; **revised** 25 June 2025; **accepted** 15 August 2025
© The Author(s) 2025. Published by Oxford University Press on behalf of the Institute of Chartered Foresters. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

mechanisms, followed by fine root mortality, which reduces the ability of the tree to provide water and nutrients to its organs. Consequently, the conditions for conducting photosynthesis no longer occur, and the tree dies. After the green attack stage, the needles turn red (*red attack*) and are finally shed (*gray attack*).

The subdivision into green, red, and gray attack, however, is not necessarily relevant for forest managers. Since the new filial generation swarms out to infest new trees ~6 to 10 weeks after the initial infestation (depending on environmental conditions, tree health, etc.) (Kautz et al. 2024), an early detection (before the emergence of the new beetle generation) is crucial to apply efficient mitigation measures such as sanitation felling and log removal. Thus, the time frame for a successful early detection in the sense of forest management usually ends 10 weeks after infestation, while the trees are typically still in the green attack stage.

The urgently needed monitoring of early bark beetle infestations has been relying on regular field visits, but also remote sensing approaches have been explored. On the ground, sniffer dogs can be trained to smell the pheromones excreted by the beetles during tree colonization (Vošvrdová et al. 2023). Additionally, trained human observers can identify infested trees by observing boring dust, entrance holes, woodpecker-caused loosened bark patches, and the beetles themselves (Kautz et al. 2023). While these ground observations achieve satisfactory performance in experiments (Kautz et al. 2023), this method is labor-intense and hardly feasible to be applied systematically across wide areas.

As opposed to that, remote sensing is a good option to monitor large areas efficiently. As stated above, the trees are deprived of water supply during the green attack stage (Kautz et al. 2024). This inevitably leads to water stress symptoms such as a reduced leaf water content and is followed by deterioration of the cell structure (Kautz et al. 2024). While early symptoms usually do not show a strong spectral response in the human-visible wavelengths (hence the term green attack), they are often detectable in wavelengths captured by multi- or hyperspectral sensors (Abdullah et al. 2018) such as multispectral satellites (Abdullah et al. 2019). For instance, satellite bands in the shortwave infrared (SWIR) wavelength range are usually sensitive to water content of the vegetation's leaf tissue (Ustin et al. 2004, Kokaly et al. 2009) and wavelengths in the red-edge and near infrared (NIR) range respond to changes in leaf cell structure.

Consequently, many remote sensing studies have shown the spectral separability between undisturbed and infested (green attack) spruce trees. For instance, Ortiz et al. (2013) use data from the multispectral RapidEye satellite with a spatial resolution of 6.5 m to distinguish between undisturbed and green attack spruce trees ~8 weeks after infestation. Another study uses WorldView-2 with a spatial resolution of 2 m to classify undisturbed and green attack trees (Immitzer and Atzberger 2014). Many studies also tested the capability of Sentinel-2 (S2) for early detections of bark beetle infestations (e.g. Bárta et al. 2021, Huo et al. 2021, Dalponte et al. 2022, and Mandl and Lang 2023).

While most of these studies indicate measurable spectral differences between green attack and undisturbed trees, the model performances achieved so far are not yet sufficient for an efficient monitoring (Kautz et al. 2024). On top of that, most studies so far focused on small study areas (e.g. Abdullah et al. 2019, Bárta et al. 2021, Mandl and Lang 2023), sometimes a single forest stand and a single image acquisition (e.g. Ortiz et al. 2013, Minařík et al. 2021). Therefore, it is unclear if the obtained results can be generalized across larger regions for a large-area application. Additionally, most of these studies use data from commercial satellites (e.g. RapidEye in Ortiz et al. 2013, WorldView-2 in Immitzer and Atzberger 2014), which are associated with costs, or exploit UAV acquisitions (e.g. Minařík et al. 2021), which are laborious to collect. Regarding freely available multispectral satellite imagery, the best current option is S2 from the European Space Agency due to its high spatial resolution of 10 m and a revisit time of ~5 days.

The temporal resolution of satellite systems such as S2 allows for a monitoring approach using time series anomaly detection. Yet, the temporal trends of satellite time series of forests can be complex due to diverse physiological responses of the trees to a multitude of environmental influences (e.g. disturbances, meteorological anomalies, environmental conditions, and stand properties), which in turn manifest in different spectral responses. On top of that, satellite time series are often noisy, e.g. due to atmospherical effects, which can lead to false detections. Consequently, a simple threshold reflectance value for anomaly detection cannot distinguish undisturbed from disturbed forest. Therefore, more advanced methods have been developed for anomaly detection using satellite time series, e.g. CCDC (Zhu and Woodcock 2014), its successor for near real-time (NRT) monitoring S-CCD (Ye et al. 2021), and BFAST Monitor (Verbesselt et al. 2012). While having shown promising results in land cover change monitoring, these methods rely on predefined functions or statistical assumptions about the time series. For instance, they assume that the processes affecting the spectral time series can be described by linear and/or seasonal trends. Consequently, they might not fully capture the complex, nonlinear relationships characteristic of ecological processes.

Studies specifically focusing on early detection of bark beetle infestations using S2 time series suggest that it is possible to some extent. However, their results cannot be directly applied in an operational system. Some studies, for instance, use information from the end of the study period to investigate detection capabilities in retrospect (Dalponte et al. 2022, Mandl and Lang 2023), which prevents NRT monitoring. Other studies investigating the early detection of bark beetle infestations on satellite time series use advanced Machine Learning (ML) techniques to improve classification of undisturbed and infested pixels (Bárta et al. 2021, Huo et al. 2021, Dalponte et al. 2022). While these models, e.g. Random Forest (RF) or Support Vector Machines, are generally capable of processing feature vectors derived from multitemporal observations, they do not inherently capture the temporal dependencies and long-range temporal dynamics of satellite time series. In other terms, they treat each feature independently and are not designed to capture that an observation at timestep t chronologically follows the observation of timestep $t-1$.

Deep Learning (DL), a sub-class of ML, has evolved as a promising tool in remote sensing of forests (e.g. Schiefer et al. 2020, Minařík et al. 2021, Schwarz et al. 2024). Recent studies also indicate its high potential in time series analysis, e.g. by forest disturbance classification (Du et al. 2023, Perbet et al. 2024, Schiller et al. 2024) and standing deadwood estimation (Schiefer et al. 2023). However, NRT monitoring that aims at early detection of disturbances in forest remote sensing using DL models has still not been tested widely. One NRT approach is anomaly detection using time series reconstructions. Here, we propose a model to reconstruct time series of S2 pixels representing undisturbed forest exclusively. Being fitted on undisturbed forest's time series only, the model is unable to reconstruct a time series of a disturbed forest's pixel, which leads to a difference between reconstruction and observation. This anomaly can be interpreted as a deviation from the undisturbed state, meaning that it must be a disturbed forest's pixel. While this approach has been tested

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

Table 1. Meta information about the datasets used in this study

Dataset	Region	Training/ test	Labels	Number of samples	Undisturbed/ infested	Max. time frame
Own work: AUS	Tyrol, Austria	Training	Visual interpretation of orthophoto	5051	Undisturbed	[2018, 2021]
Own work: BB	Brandenburg, Germany	Training	Visual interpretation of orthophoto	13 041	Undisturbed	[2018, 2021]
Own work: SAX	Saxony, Germany	Training	Visual interpretation of orthophoto	13 760	Undisturbed	[2018, 2021]
Own work: THU	Thuringia, Germany	Training	Visual interpretation of orthophoto	11 712	Undisturbed	[2018, 2021]
Own work: RLP	Rhineland-Palatinate, Germany	Training	Visual interpretation of orthophoto	963	Undisturbed	[2017, 2020]
FNEWS LoSa	Lower Saxony, Germany	Training	Visual interpretation of orthophoto	22 186	Undisturbed	[2016, 2018]
FNEWS Sax	Saxony, Germany	Training	Visual interpretation of orthophoto	12 073	Undisturbed	[2016, 2019]
FNEWS BW1	Baden-Württemberg, Germany	Training	Visual interpretation of orthophoto	95 560	Undisturbed	[2016, 2017]
FNEWS BW2	Baden-Württemberg, Germany	Training	Visual interpretation of orthophoto	61 448	Undisturbed	[2016, 2019]
Lux	Luxembourg	Training	Visual interpretation of orthophoto	96 159	Undisturbed	[2016, 2019]
AF (Arnsberg Forest)	North Rhine-Westphalia, Germany	Test	Ground observation	4939 (undisturbed) 351 (infested)	Undisturbed, infested	[2017, 2021]

Max. time frame refers to the years of Sentinel-2 data contained in the time series of the corresponding dataset.

successfully in anomaly detection in other fields of research (Wu et al. 2023), it is still novel in remote sensing to the best of our knowledge.

Accordingly, in this study we tested a reconstruction-based DL anomaly detection method on S2 time series for its capability of early detections of bark beetle outbreaks in spruce forest in Central Europe. To this aim, we adjusted a state-of-the-art model architecture to be able to cope with irregular time series and thus avoid interpolation and resampling. We acquired >300 000 S2 time series of undisturbed forest in Germany, Austria, and Luxembourg for model training. We trained our model on training sites located far away from the test study site in order to investigate if the approach can be generalized across regions. The latter is still an open challenge and a strong limitation of most former studies on early bark beetle infestation detection (Marvasti-Zadeh et al. 2023). Additionally, most other studies focus on detecting the green attack stage after infestation, which is not relevant for forest management (see above). Here, we explicitly investigate how many weeks after infestation are needed to detect a bark beetle infestation. In case of adequate performance, such a model could readily be applied to Central European spruce forests using the freely available S2 data to finally enable an operational NRT monitoring system of early bark beetle infestations.

Methods

Training dataset preparation

In this study, we use pixels representing undisturbed forest for training and pixels representing both undisturbed as well as bark beetle–infested forest for testing (for an overview, see Table 1 and Fig. 1). We compiled multiple datasets from different publications and sources for model training. These comprise data from Langner et al. (2023) (in the following "FNEWS"), from Schwarz et al. (2024) (in the following "Lux"), and from Schiller et al. (2024) (in the following "own work": RLP, BB, THU, SAX). The first author of the present paper additionally labeled one study site in Austria ("own work": AUS). All of these datasets contain forest disturbance information. In case of our own work (RLP, BB, THU, SAX, AUS), we removed the forest disturbance areas (dieback and logging) including a buffer of 15 m from all forested areas. Lux contains visually interpreted dieback areas and no forest mask. Here, we utilized the official forest land use as well as dieback and logging information from the Cadastral and Topographic Administration (Administration du cadastre et de la topographie, ACT; www.geoportail.lu) of Luxembourg including a buffer of 15 m to remove non-forest as well as disturbed (logging and dieback) areas. FNEWS comprised only forest areas, so a forest mask was not necessary. Here, we discarded all but the polygons denoted as undisturbed. We further discarded all but the coniferous forest pixels from all datasets as determined by the forest type characterization of the Copernicus Land Monitoring Service or, in case of Lux, by the official forest type information from the Cadastral and Topographic Administration of Luxembourg. All S2 pixels located entirely within the remaining coniferous forest areas were assumed to show time series of undisturbed forest and, thus, their time series were sampled as training data from the S2 datacube as described in Section Time Series Acquisition.

Test dataset preparation

For model testing, we needed spatially and temporally explicit data about bark beetle infestations as well as a control group (undisturbed areas). The former was obtained from Wald und Holz NRW (North Rhine-Westphalia; in the following "AF"; data collector: J.M.) in or near Arnsberg Forest in North Rhine-Westphalia, Germany, within the scope of the FirSt 2.0 project. AF contains both undisturbed and infestation information, all infested and undisturbed data being located within a range of 45 km from the city of Arnsberg (Fig. 2).

The 78 undisturbed polygons consist of monocultural stands of Norway spruce that have been confirmed to be undisturbed and green by 14 June 2021 by the local foresters. For the infested polygons, ground observations were conducted every 2 weeks within the study period in the Arnsberg Forest in an area of ~2.5 × 2.5 km. In this area, only a few spruce stands were still alive and uninfested in 2021. Thus, it was feasible to scan every single spruce tree for signs of bark beetle infestations in the study period. After detection of these signs (e.g. saw dust, bore holes, developmental stage under bark, beetles themselves), areas of similar developmental stage of the bark beetles were determined by screening each tree. The polygons delineating the canopy areas of the infested trees were derived using in-field GPS measurements and delineated and checked using GIS software. The infestation

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

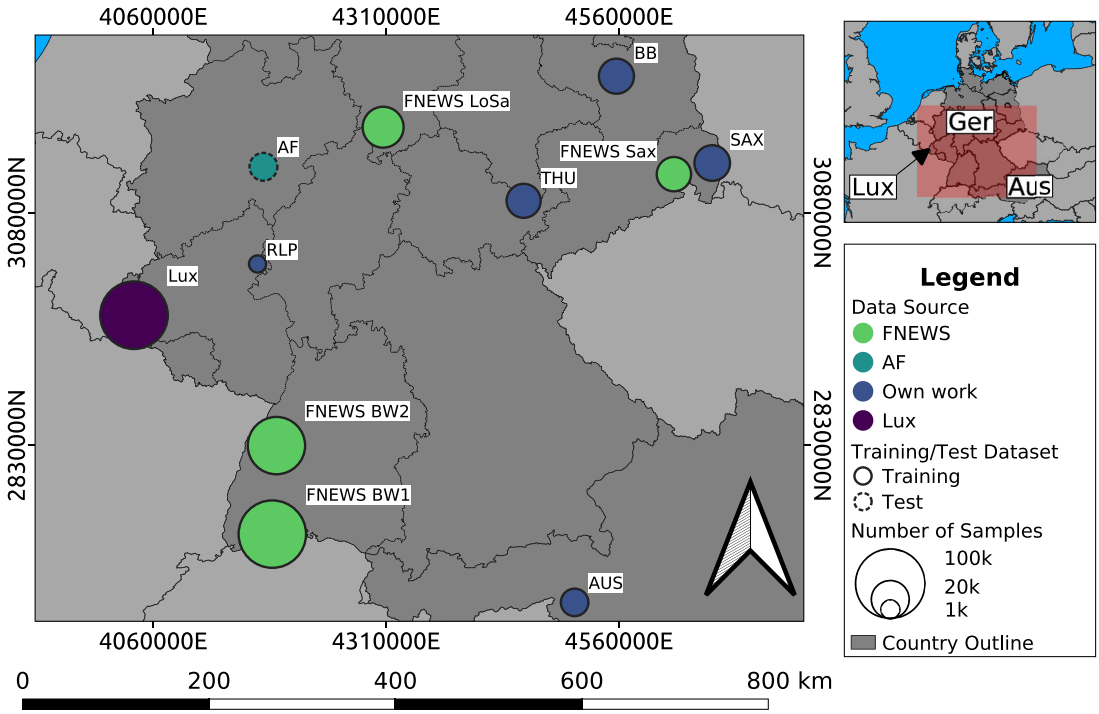


Figure 1. Spatial distribution of datasets used in this study, spanning the countries Luxembourg (Lux), Austria (own work), and Germany (own work, FNEWS, AF). Center point of each circle on the map is located at the centroid of the corresponding dataset. Training datasets are indicated by solid, test dataset by dotted outline. Map on the top right shows an overview of the study countries (Lux: Luxembourg; Ger: Germany; Aus: Austria). Point sizes are scaled proportionally to the square root of the number of samples to accurately represent data magnitude while accounting for perceptual bias. Coordinate reference system EPSG:3035.

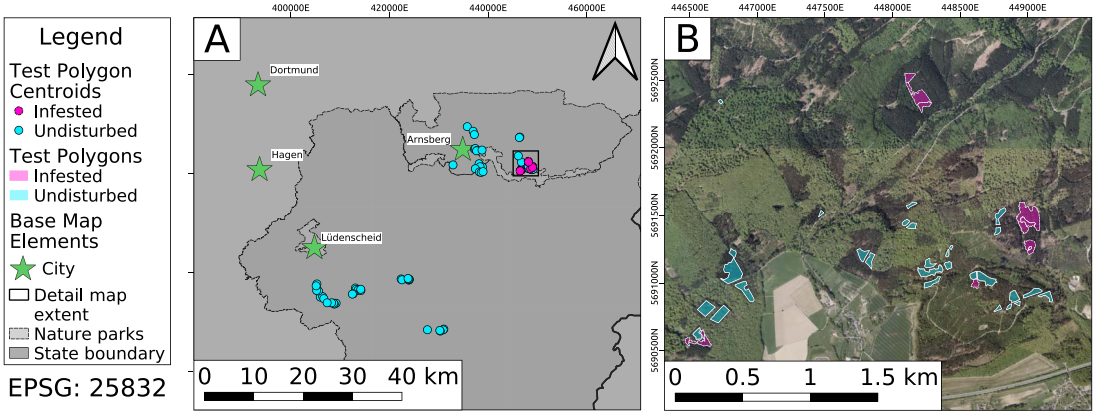


Figure 2. Overview of test area (A) and detailed map of infestation monitoring area (B) of the test dataset AF. For orientation, larger cities are shown by star symbols and nature parks are visualized by polygons (Arnsberg Forest in the north and Sauerland-Rothaargebirge in the south). Coordinate reference system: EPSG:3035. Digital orthophotos courtesy of open.NRW (dl-zero-de/2.0).

date could then either be inferred directly (e.g. when the bark beetles were freshly boring into the bark; 6 polygons) or estimated depending on the detected developmental stage of the bark beetles for certain areas (5 polygons). Thus, the temporal information of the bark beetle infestations could be inferred precisely, likely achieving an uncertainty of only a few days. In the following, we assume the day inferred by this procedure based on the field visits as the infestation date. For this study, we used only those polygons that have not been harvested until 14 weeks after the presumed infestation date, as including clearings would likely lead to an overestimation of model performance. We also excluded all S2 pixels that were not fully covered by the disturbance polygons. Details about the undisturbed and infested datasets of AF are provided in Table 2. As visible in Fig. 1, training and test data do not overlap and are far away from each other, enabling an external validation that investigates the model's capability of generalization across regions.

Time series acquisition

For each S2 pixel sampled as described in the previous sections, we acquired a time series from an S2 datacube preprocessed with the Framework for Operational Radiometric Correction for Environmental Monitoring (FORCE; Frantz 2019). The Level 2A data derived from FORCE have been atmospherically and topographically corrected and the two S2 satellites S2A and S2B have been co-registered. The latter is crucial for time-series analysis as it is necessary to ensure that the reflectance of the same geolocation is sampled at each timestep. Otherwise, the concept of

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

Table 2. Information on AF dataset used for testing, including number of polygons and pixel count of both undisturbed and infested datasets as well as infestation date range of infested dataset

Subset	Number of polygons	Number of pixels per polygon (min; mean; median; max)	Infestation date range
Undisturbed	78	1; 63; 29; 603	
Infested	11	1; 32; 19; 155	28 May 2021 to 3 August 2021

change detection would not be feasible. Furthermore, the FORCE Level 2A product has been cleared of clouds, cloud shadows, snow, and other potential artifacts. All S2 bands have been resampled to 10 m spatial resolution using the *ImproPhe* algorithm implemented in FORCE. The datacube for Germany was readily available on the platform www.eo-lab.org, while the datacubes for Austria and Luxembourg were processed using the same FORCE parameters using our own computational resources.

As we pursue NRT monitoring, high-quality data are of utmost importance since any low-quality observation such as a cloud that has been missed by the cloud masking algorithm could potentially lead to an anomaly, which then triggers a false warning. Prioritizing quality over quantity, we used the quality assurance information from FORCE to further exclude observations from analysis when flagged as cloud (opaque, cirrus), cloud shadow, snow, water, high or unknown aerosol optical depth, and others. FORCE uses an improved *fmask* algorithm to detect clouds (Frantz et al. 2018).

The model was provided with univariate time series of the CR_{SWIR} (continuum removal shortwave infrared) VI as input. It is computed using the narrow NIR (B8a), SWIR1 (B11), and SWIR2 (B12) bands of S2. The equation to compute CR_{SWIR} is

$$CR_{SWIR} = \frac{B11}{B8a + \left(\frac{B12-B8a}{\lambda_{B12}-\lambda_{B8a}}\right) \times (\lambda_{B11} - \lambda_{B8a})} \quad (1)$$

where λ_{Bn} denotes the S2 wavelength of the corresponding band in nanometers and Bn represents the observed reflectance of the corresponding S2 band. Using this equation, a straight line (the *continuum*) is drawn between the NIR and SWIR2 band. Next, the SWIR1 reflectance is compared to the continuum value at the SWIR1 wavelength. Thus, the deviation of SWIR1 to this continuum is measured, specifically highlighting the reflectance of the forest in the SWIR1 spectral range, which is sensitive to changes in water content of the vegetation (Ustin et al. 2004, Kokaly et al. 2009). Since reduced leaf water content is an early symptom of bark beetle infestations in spruce trees and SWIR1 is sensitive to vegetation moisture and bark beetle green attacks (Abdullah et al. 2018), the CR_{SWIR} index is particularly suitable for early detection of bark beetle infestations (Dutrieux et al. 2021). We also tested other VIs suggested by, for instance, Abdullah et al. (2019), König et al. (2023), and Mandl and Lang (2023). As they resulted in much lower model performance, we do not report their results in this study.

All datasets used in this study are temporally explicit, meaning that the date of image acquisition and the date of ground observation of the bark beetle status of the trees were both known for the training and validation data, respectively. This was necessary to allow the acquisition of undisturbed (and non–green attacked) time series in training and validation, and to enable measuring the detection delay. The time series were acquired according to the following scheme: for undisturbed pixels, the end date was defined as 31 December of the year prior to the observation date, while for infested pixels, the end date was set to 31 December of the same year. As infested spruce trees usually express mortality within 1 year, this guaranteed that the time series did not contain an early stage of bark beetle infestations that was not yet visible in the orthophoto (green attack). The dates of image acquisition for the training data can be inferred from the original publications, see Schiller et al. (2024) and Langner et al. (2023). The maximum length of the time series was set to four years and only full years were used, meaning that the time series always started at the first observation of the corresponding Gregorian calendar year. The earliest start date was set to 1 January 2016, as 2016 is the first year with S2 coverage from the beginning of the calendar year. Thus, the resulting time series exhibited a length between two and four years (cp. Table 1).

Reconstruction-based time series anomaly detection

There are multiple ways to approach NRT change detection in time series using DL, e.g. short-term forecasting, meaning that the model predicts future observations, which are then compared to real observations once the latter are acquired. In this study, we tried a reconstruction-based approach that works as follows: the model is trained to reconstruct a given (here: univariate) time series, i.e. it attempts to learn the patterns underlying each of the time series and then reproduces them as close as possible, optimized by a loss function (here: mean squared error between observations and reconstructions). Therefore, the input and output of the model is a time series: input are the observations, output are the reconstructions/predictions.

During the training phase, the model is only exposed to non-anomalous time series, derived from pixels of fully undisturbed forest in our case. Thereby, the model learns patterns (*representations*) of undisturbed time series. As opposed to that, during the test phase the model receives both anomalous and non-anomalous time series as input. As the model learned only patterns from S2 time series of undisturbed forest, it would be unable to reconstruct disturbed (anomalous) time series. Consequently, the reconstructions and the observations diverge beginning at the timestep at which the anomaly starts. This discrepancy can be measured and classified as an anomaly if exceeding a user-defined threshold.

TimesNet model

In this study, we used the anomaly detection module of the *TimesNet* model (Wu et al. 2023) to reconstruct the input time series. TimesNet has shown state-of-the-art performance on benchmark datasets across all time series analysis tasks, including anomaly detection (Wu et al. 2023). The principle of this model architecture, which uses the concept of *Convolutional Neural Networks* (*CNNs*) as main building block, along with our adjustments to it are explained in the following.

First, the input time series are embedded into a latent space by two embedding modules: a 1D-convolutional layer provides links between the input channels, i.e. across the channel dimension

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

and ignoring the time dimension (*Input Embeddings*). In our case, the only input channel is a time series of CR_{SWIR}. Therefore, the 1D kernels translate the CR_{SWIR} observations into the *latent space* of the model, learning representations relevant to the understanding of the undisturbed time-series patterns. In the case of multivariate time series, the kernels would additionally model input channel combinations relevant for the understanding of the time series. Additionally, *Temporal Embeddings* (also called *Positional Encodings*; PE) are derived as learnable sine and cosine representations from the day-of-the-year (DOY) information of the observations and summed to the Input Embeddings. The anomaly detection module of the TimesNet architecture is only designed for and tested on regular time series (with equidistant timesteps), and thus uses the position of the observation within the sequences (the *index*) for PE. Since S2 time series are usually irregular (due to pixels affected by clouds, cloud shadows, etc.), we adjusted the PE to take as input the DOY values of the input time series. Former research has shown that DL models can learn PE on DOY values that support understanding the notion of time of each observation (Yuan and Lin 2021, Wu et al. 2023, Schiller et al. 2024).

The idea of the TimesNet model is to learn representations across and within temporal subsets of the input time series (inter- and intra-period variations; Fig. 3 (1) and (2)) by splitting them into sub-sequences of repeating temporal patterns (*periods*) determined by Fourier transform. In our case of highly irregular time series, however, a Fourier transform is not applicable. To mitigate this, we use domain knowledge to replace it: the most relevant period in forest remote sensing covering several years is the yearly cycle of growing season and non-growing season. Thus, we can define the period as one year. Technically, our input time series (of two to four years' length) is split by years and concatenated next to each other instead of being analyzed as one single time series of two to four years' length (Fig. 3 (2)). With this setup, however, the *convolutional kernels* (the main building blocks of CNNs) running across the years (inter-period variation) would still have to compare observations of potentially very different time steps, as these kernels technically stride over index positions of the sequences (Fig. 3 (3)). The abovementioned PE mitigate, but do not solve, this problem, as e.g. the first observation of each single year from the multi-year time series can be >2 months apart. For instance, the observation date of the first index position of a year (the first observation of this year) might be 1 January 2016, whereas the observation date of the succeeding year's first index might be 20 March 2017. Since PE technically alters the value of the observation in the latent space of the model given the day-of-the-year value, but do not consider the temporal context of each observation with respect to its preceding and succeeding observations (which is exactly what convolutional kernels are supposed to do), a more flexible approach that considers temporal context of the utilized convolutional kernels is needed. Therefore, we adopt the concept of *Time-Parameterized Convolutional Neural Networks* (*TPCNNs*) (Kosma et al. 2023) by introducing "time-aware" kernels with learnable parameters dynamically derived from the DOY values of the corresponding observations (Fig. 3 (4)). Technically, the learnable weights of the convolutional kernels are multiplied by the time tensor. Other than in the TPCNN study, we use absolute (and not relative) parameterization. Relative parameterization means that the center weight of the convolutional kernel is not altered by the temporal value (the timestamp), while the others are adjusted in relation to the center value. This, for instance, might be reasonable for tracking the spread of a wildfire captured by a sequence of satellite images, where the absolute timestamp (e.g. the date) is irrelevant. In our case, however, the absolute DOY contains essential information, e.g. the temporal position in the vegetation season (summer, autumn . . .) of the observation. Consequently, we use the DOY of each embedded observation to introduce dynamic time-aware convolutional kernels. Instead of the DOY itself, we use the sine value of the DOY to model the seasonal cycle. This allows the model to understand that while DOY 1 and DOY 365 are mathematically the farthest apart in the sequence, they are almost identical in terms of the seasonal cycle. The convolutional kernels used in TimesNet always cover all sequences at once, meaning that their width is equal to the maximum number of parallel sequences, which is four years (Fig. 3 (3)). In case the time series contains less than four years, the years without observations are padded with zeros for technical reasons. The kernels vary in their temporal coverage from one until 15 time steps (this number was determined to be best in preliminary training runs). Thus, these dynamic kernels enable modeling the intra-period variation (see above) as well. Using the representations learned as described above, the model then reconstructs the input time series using a *multi-layer perceptron* (also called *dense layer*; Fig. 3 (5)), which is the most basic layer in DL.

For details about the model parameters and exact implementation, refer to the code repository on https://github.com/ChrSchiller/TimesNet_RSAD.

Training and test procedure

We trained the model on all univariate CR_{SWIR} time series from the training datasets (Table 1; Fig. 1) of two to four years' length. We used data augmentation to inflate the training dataset during training in order to prevent overfitting of the model to its specific characteristics. In this approach, we applied DOY jitter, which randomly adjusts the DOY values of the observations by ±3 days, and observation jitter, which modifies the satellite observations by up to 2% of the standard deviation of the corresponding time series. The jitter was intentionally small since our aim in this study was to identify even small diversions from normal patterns. Thus, the noise level should be as low as possible in the time series. As an additional data augmentation procedure, we shuffled the order of the years in the time series randomly. This ensured e.g. that there is no unintended bias from the more limited time series of 2016 and 2017 (when S2B was either unavailable or only partly available), which would otherwise always be the earliest years in the time series. We stopped the training procedure if the model performance (loss determined as mean squared error) did not improve for three consecutive training cycles (*epochs*). This procedure is called *Early Stopping* with a *patience* of three and further prevents overfitting.

We split the training dataset randomly by a ratio of 4:1 into training (used for adjusting the model parameters during training) and validation. Mean and SD of the training dataset were used to normalize the validation and test data. We repeated the random training data split three times to assess whether the model's performance is consistent across different training datasets. Note that when referring to the validation dataset here, we do not mean the dataset used for validation of the model. While the latter is referred to as the test dataset, the validation dataset is used to track the progress of the training process in DL.

Analyses

The predictions of the final model on the validation dataset were used for automatic thresholding for the classification on the test dataset by determining the background noise level of the model.

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

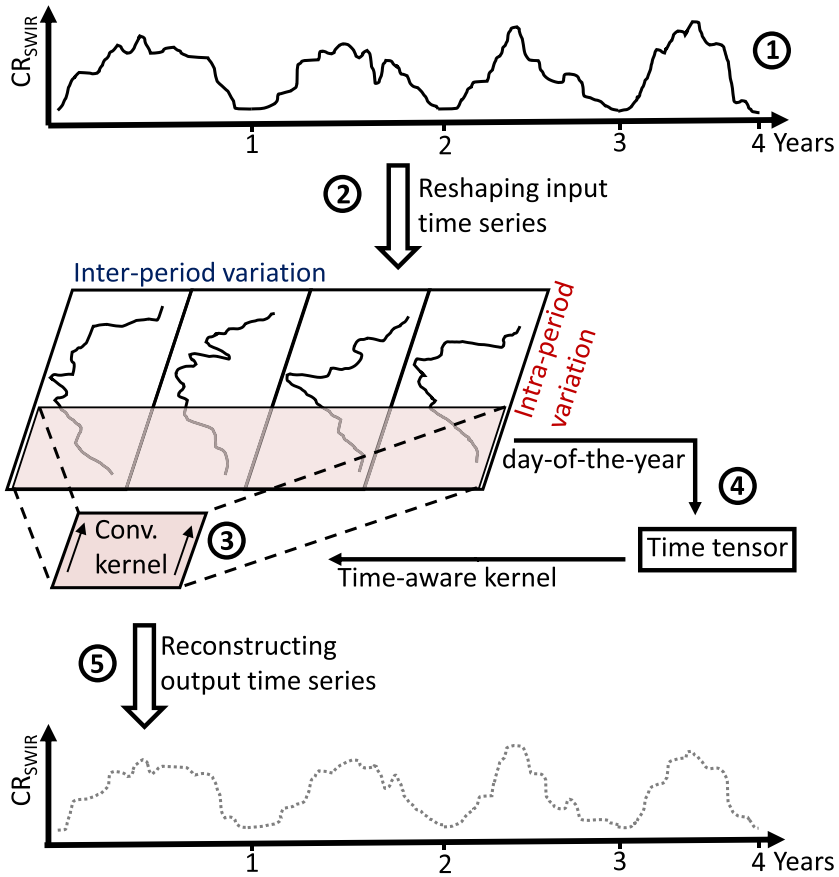


Figure 3. Processing chain of our adjusted version of the TimesNet (Wu et al. 2023) model architecture. (1) The input time series of the CR_{SWIR} index of up to 4 years' length are (2) reshaped by splitting them by years. Afterwards, (3) convolutional kernels stride over the time series to learn representations across sequences (inter-period variation) and timesteps (intra-period variation). The convolutional kernels are (4) dynamically parameterized by a time tensor (Kosma et al. 2023) consisting of adjusted day-of-the-year values of the corresponding observations. (5) The learned representations are used to reconstruct the input time series as close as possible.

The decision threshold for a bark beetle infestation detection was defined as

$$\text{threshold} = \text{mean}_{\text{val_error}} + f \times (\text{q99}_{\text{val_error}} - \text{q01}_{\text{val_error}}) \quad (2)$$

where $\text{mean}_{\text{val_error}}$, $\text{q99}_{\text{val_error}}$, and $\text{q01}_{\text{val_error}}$ denote the mean, 99th, and 1st percentile of all the model errors (squared error between normalized observation and reconstruction) of the validation dataset, respectively, and f is a *threshold factor*. The inter-percentile range was preferred over the standard deviation here to prevent rare extreme outliers from influencing the threshold. Three threshold factors were tested in this study: 0.5, 1, and 1.5. An anomaly was detected if the squared error between reconstruction and corresponding normalized observation exceeded the determined threshold for two consecutive observations. We removed non-growing season observations (between 15 November and 15 March) from analysis since bark beetle infestation detections are not possible in this time due to the inactivity of the beetles in winter and might lead to unintended false warnings.

We evaluated the model performance on undisturbed time series of the AF dataset between 1 January 2018 and 31 December 2020 and on infested time series between 1 January 2019 until 10 to 14 weeks after the infestation event in 2021. Using this approach, the time series of both groups had a length of exactly three years for the undisturbed class and almost three years for the infested class. A detection was considered a true positive only if a warning was triggered according to the above-mentioned scheme within the time frame of 4 weeks before and 10 to 14 weeks after the infestation (in the following called "detection period"). We analyzed the early detection capabilities of the model by consecutively increasing the detection period by 1 week from 10 to 14 weeks after the infestation. If a warning was triggered before the detection period in the infested time series (meaning between 1 January 2019 and the corresponding infestation date in 2021), we added a false positive to the confusion matrix for each warning issued too early by the model. Regarding the undisturbed time series, we decreased the number of true negatives and increased the number of false positives by one for the first (false) warning, and added a false positive to the confusion matrix for any further false detection per pixel. The confusion matrix was used to compute producer's accuracy (PA, syn. recall), user's accuracy (UA, syn. precision), and F1-score (harmonic mean between PA and UA). These metrics were used to assess the performance of the model.

We used R programming language (R Core Team 2023, v4.4.2) including the packages terra (Hijmans 2023, v1.7-83) and sf (Pebesma 2018, v1.0-19) for preprocessing and result analysis and Python (Van Rossum and Drake 2009, v3.8) including PyTorch (Paszke et al. 2019, v2.2.2) for model training and inference.

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

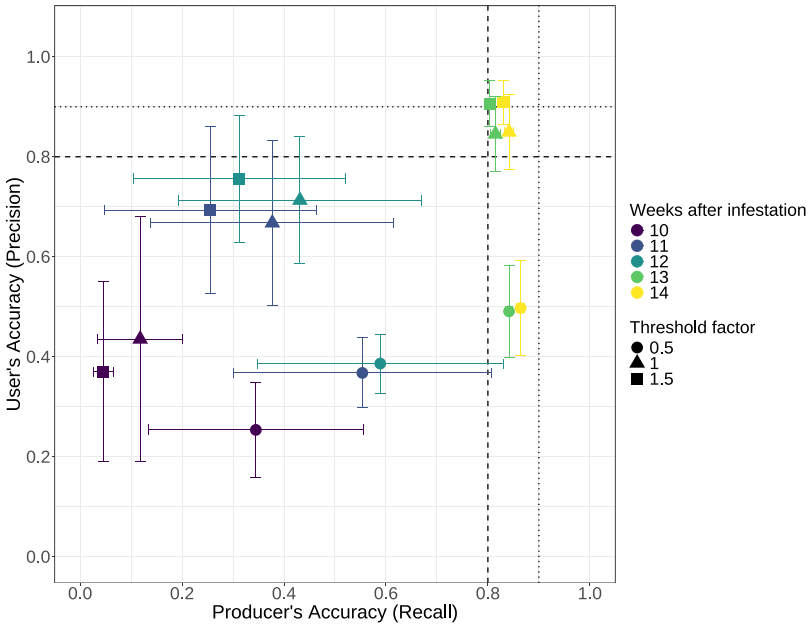


Figure 4. Mean and standard deviation of user's accuracy (precision) and producer's accuracy (recall) for threshold factors (symbols) of decision threshold (for details see section Methods) and weeks after infestation allowed for detection (colors) of three random splits of training and validation datasets for model training. Dashed (dotted) black line indicates value of 0.8 (0.9) for reference.

Results

Using a threshold factor of 1, the model achieves a PA of 11.8% ± 8.4% and a UA of 43.5% ± 24.5% (Fig. 4, Supplementary Tables S1 and S2) in the early detection scenario (detection within 10 weeks of estimated/observed infestation). This results in an F1-score of 0.18 ± 0.12. When allowing for a detection within 13 weeks, both PA and UA increase notably to PA = 81.5% ± 1%, UA = 84.5% ± 7.6%, and F1-score = 0.83 ± 0.04 (Fig. 4; Supplementary Tables S1 and S3). For detailed results on the detection performance after 11, 12, and 14 weeks and the other threshold factors, see Supplementary Tables S1, S4, S5, and S6. In case of a threshold factor of 0.5 in the early detection scenario, the UA is strongly reduced with UA = 34.5% ± 8.4%, whereas the PA is clearly increased with PA = 25.4% ± 9.4%. Here, the F1-score is 0.29 ± 0.14.

Among the 351 time series labeled as infested, 60 are detected in the 6th week after infestation in case of a threshold factor of 1 and the best model resulting from the three training data splits (Fig. 5), whereas 59 are not detected within the detection period. Most of the detections occur in week 11 ($n = 123$) and week 13 ($n = 65$). As visible in Fig. 6, detections with a similar delay in weeks after infestation cluster spatially (another example is shown in Supplementary Fig. S1). In this example, the center of the disturbance polygon gets detected first (maximum 10 weeks after infestation), and those pixels located more toward the edges of the polygon get detected later (later than 10 weeks after infestation). An example time series of the abovementioned polygon's pixels shows that the anomaly score rarely supersedes the detection threshold in case of a threshold factor of 1 (Fig. 7). In this example, it occurs only once before the infestation date. This occurrence would not trigger a warning since the detection threshold is only superseded once. Apart from that, the model reconstructs the time series accurately. Shortly after the infestation, however, the observed CR_{SWIR} time series deviates from normal, which manifests in a strong and rapid increase of the anomaly score. In this case, subtle deviations from the non-anomalous pattern result in an early detection.

Discussion

In this study, we used >300 000 Sentinel-2 (S2) time series derived from pixels of undisturbed coniferous forest in Central Europe (Germany, Luxembourg, Austria) to train a reconstruction-based Deep Learning (DL) model for anomaly detection with the goal of early detection of bark beetle infestations.

We quantified differences between observations and reconstructions (which are our predictions) of the CR_{SWIR} index to detect spectral anomalies resulting from bark beetle infestations. While our model showed only a few detections within 10 weeks after infestation (early detection scenario), which would have been preferred by forest managers to prevent the spread of the next beetle generation, it achieved high performance metrics within 13 weeks after infestation (PA = 81.5% ± 1%, UA = 84.5% ± 7.6%, F1-score = 0.83 ± 0.04). In the following, we discuss the model performance and the model's potential in operational monitoring.

Model performance

Using a threshold factor of 0.5 resulted in many false positives (Fig. 4), showing that this decision threshold is not clearly exceeding the background error of the reconstructions. Even using this threshold, however, the PA, which is a measure of the ability of the model to detect positive cases (infestations), was only 34.5% ± 21.1% in case of the early detection scenario. The high standard deviation of UA and PA in the early detection setup independent of the threshold factors also indicates that the results are not robust for early detection. In any case, the model clearly failed to reach a PA of 80% or higher, which was stated to be supportive of an operational system aiming at controlling the number of bark beetle infestations (Pietzsch et al. 2023). Figure 7 illustrates why the early detection results might be unstable: while the model detects a subtle anomaly as the CR_{SWIR} value decreases unexpectedly in July 2021 in this example, the observed CR_{SWIR} strongly increases afterwards, leading to a clear anomaly. The latter spectral response is expected once the trees are deprived of water supply, which is a typical early symptom of bark beetle infestations (Kirisits and Offenthaler 2002, Kautz et al. 2024,

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

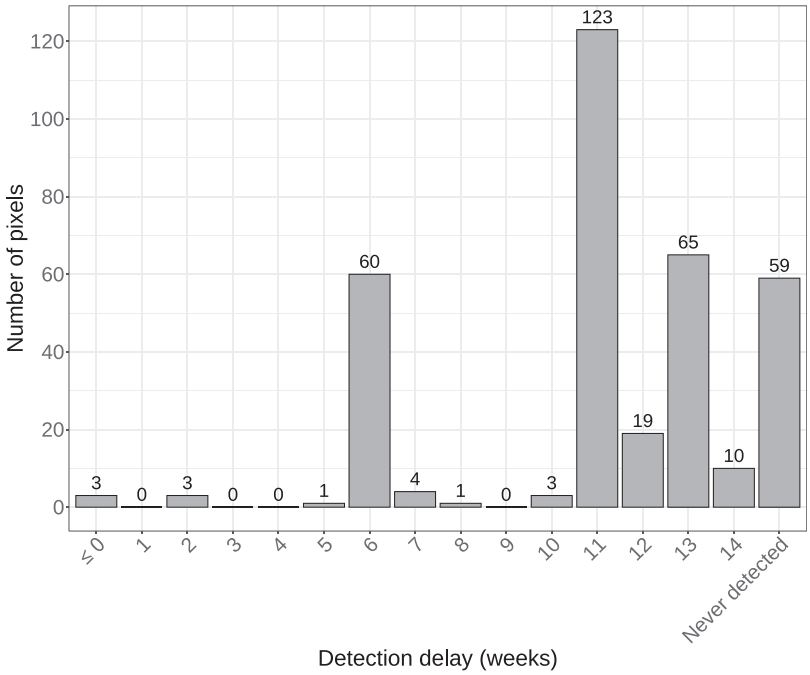


Figure 5. Histogram showing the number of pixels of infested polygons with respect to the relative week of detection after estimated/observed bark beetle infestation using the best among the three random training sample splits and a threshold factor of 1 (see section Methods). Last bin ("never detected") are those pixels that were not classified as anomalous until 14 weeks after infestation.

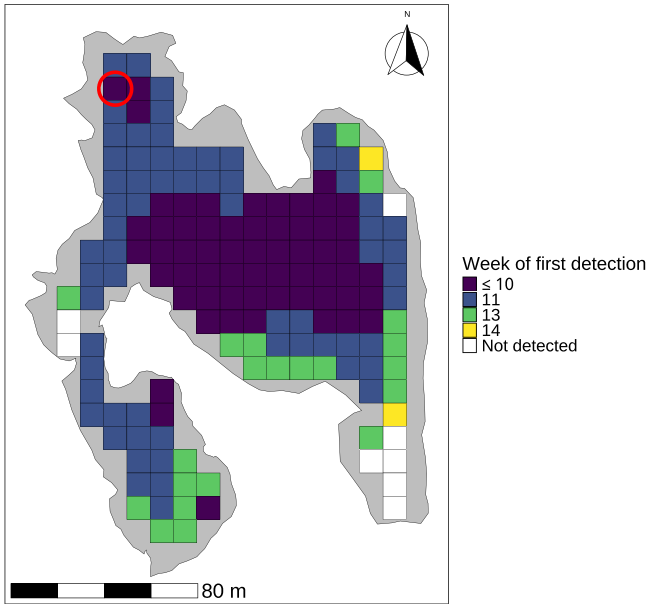


Figure 6. Example polygon of infested class and time of detection in weeks after infestation for all Sentinel-2 pixels completely covered by the polygon for the best model resulting from the three random training sample splits and a threshold factor of 1 (see section Methods). Circle: time series shown in Fig. 7. Coordinate reference system EPSG:3035.

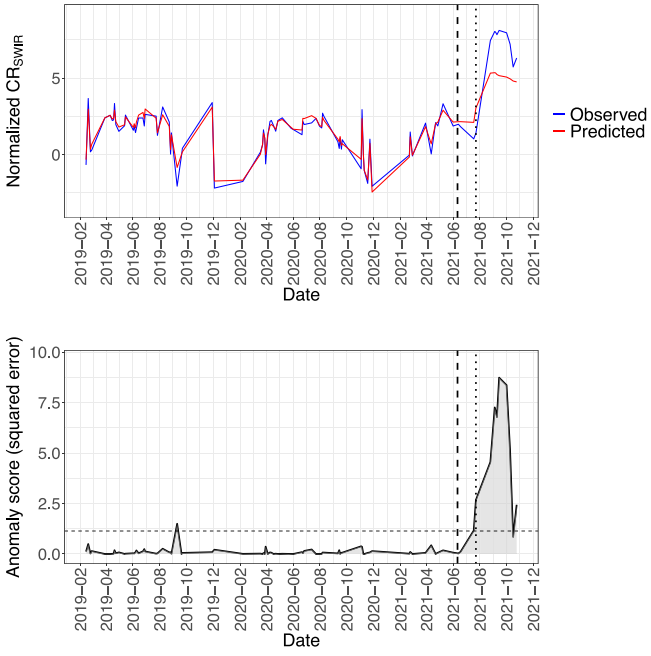


Figure 7. Example time series of observed and predicted normalized CR_{SWIR} (upper panel) of an infested forest stand's pixel and corresponding anomaly scores (lower panel) of the best model among the three random training sample splits. Vertical dashed line: date of infestation (11 June 2021); vertical dotted line: date of detection using a threshold factor of 1 (23 July 2021); horizontal dashed line: decision threshold of an anomaly score of 1.138 (threshold factor = 1; see section Methods). The plot shows the time series of the highlighted pixel in Fig. 6.

Marx et al. 2024). The former, however, is more likely an uncommon spectral fluctuation than an indicator of a typical physiological response of the trees, as this spectral behavior is, to the best of our knowledge, not described in literature. Moreover, this is inconsistent with other time-series examples (not shown for brevity). In week 11 and again in week 13, the detection rate suddenly and strongly improves (Fig. 4). From week 13 onward, the results are stable across random training data splits, indicating that our approach is robust and the model is performing well when given more time to detect the disturbance. This might be the transition from green to red attack, when the trees show a clear physiological and thus spectral response. This is in line with

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

Huo et al. (2021), who found that the spectral response of S2 only shows significant differences between undisturbed and infested trees for severely infested plots (>10 trees) and only after ~3 months, at the end or after the green attack stage.

Accordingly, it is unlikely that the correct disturbance detections in weeks ≤0 and 2 after the infestation (Fig. 5) were actually caused by the targeted bark beetle infestation. While potential reasons for these early detections could have been errors in determining the infestation date or an overseen tree infested earlier during ground sampling, it seems more likely that the bark beetle disturbance constitutes a secondary disturbance on an already declining tree. In the latter case, the detected disturbances would either be a delayed symptom of the severe hot drought of the preceding years (2018–2020) or caused by an unknown disturbance agent, whereas the bark beetle infestation itself might have followed, rather than initiated, the tree's mortality. Delayed mortality occurring after the end of a drought has been reported before (Bigler et al. 2007, Haberstroh et al. 2022). Secondary disturbance as a reason for pre-infestation detections is also supported by the fact that false warnings are much more abundant in the infested than in the undisturbed time series (cp. Supplementary Fig. S2). The figure also shows that these false warnings are accumulated by a very small amount of pixels, some of them reaching up to 16 warnings for a single pixel that was infested much later and potentially disturbed already at the time the anomaly was detected by the model (also cp. Supplementary Fig. S3). The particular example of Supplementary Fig. S3, which returned three false warnings, shows a pixel located at the boundary of the test polygon, and the neighboring forest stand has been cleared due to bark beetles in the previous year (also confirmed for most other pixels expressing more than five false warnings). Thus, it is possible that the reason for the (seemingly false and too early) warnings of such pixels is a spatial inaccuracy of the polygon delineation. Since these few pixels expressing early warnings had a large impact on our current analyses, although they might be correct detections (except that they detect another disturbance event), we propose an alternative performance assessment that defines "anomalous phases" instead of counting each two consecutive anomalous observations as a false positive in the Supplementary Material (see also Supplementary Fig. S4, Supplementary Table S7). An anomalous phase starts with the first detection and ends only if two consecutive observations do not exceed the anomaly threshold. In any case, the detections within a few weeks after infestation illustrate that the model, as all other remote sensing–based near real-time (NRT) monitoring systems we are currently aware of (e.g. Verbesselt et al. 2012, Puhm et al. 2020, Dutrieux et al. 2021, and Ye et al. 2021), is not able to differentiate between disturbance agents. One option to partly differentiate between disturbance agents would be a *post hoc* assessment investigating the date of detection and the shape of disturbance patches. For instance, windstorms and harvesting usually occur in winter in Central Europe, whereas fires usually occur in summer.

Once a disturbance is detected by the model, the anomaly often manifests unambiguously. The difference in both UA and PA is rather small for the threshold factors of 1 and 1.5, which illustrates that both decision thresholds are well above the reconstruction error of the undisturbed time series. This is supported by Fig. 7, which shows a clear distinction between anomaly scores before the infestation and when the deviation of the observations from the undisturbed pattern starts. Consequently, most false positives can be avoided by our approach by selecting a threshold factor of ≥1. Thus, we do not need to apply a minimum mapping unit of multiple pixels to avoid falsely detected anomalies of single pixels, as it is sometimes required in other NRT monitoring approaches (e.g. Puhm et al. 2020 and Reinosch et al. 2024). Additionally, the model should be able to reconstruct nonfatal drought stress and thus not trigger a (false) warning in most cases. As the spectral trajectories of trees during the hot drought of 2018 until 2020 in Central Europe have been included in the training dataset, the mentioned nonfatal stress was indeed part of the training. Thus, these severe stress events—when not leading to mortality of the dominant trees—should mostly not be considered anomalous by the model.

All in all, our study design along with the results suggest that our method can safely be generalized and thus utilized in locations other than those used for training purposes. As seen in Fig. 7 (among others), the model responds immediately once the spectral trajectory deviates from the pattern of the preceding years. Thus, the model appears to generalize well across different locations.

Operational bark beetle monitoring

Given that the model shows an immediate response once the spectral trajectory deviates from the normal pattern (visible in Fig. 7), it is questionable if a timely bark beetle infestation detection (within 10 weeks after infestation) can at all be realized with S2 time series using any model, especially in a large-area approach as we try here.

Many studies on early detection of bark beetle (mostly *Ips typographus* L.) infestations have already been conducted. They exploited S2 and other sensors such as uncrewed aerial vehicles (UAVs) or other satellites (e.g. WorldView, RapidEye). The most promising results with PA and UA >70% were achieved on individual tree level using high-resolution imagery, e.g. UAV (Minařík et al. 2021; spatial resolution 6 cm), WorldView-2 (Immitzer and Atzberger 2014; 2 m) or RapidEye (Ortiz et al. 2013; 6.5 m). Model performances using S2 are generally lower, e.g. 53%/67% (PA/UA; Abdullah et al. 2019), 76%/61% (Mandl and Lang 2023), 76%/53% (Dalponte et al. 2022), and 79%/57% (Bárta et al. 2021). This might be attributed to the lower spatial resolution of this sensor, making an analysis on individual tree level challenging (for an attempt see Dalponte et al. 2022). When counting a true positive only within 10 weeks after infestation in our analysis, all of these studies outperformed ours since it did not show a consistent response within this period (exceptions are discussed in Section Model Performance and visible in Fig. 6). However, if using a less strict criterion, e.g. detection within 13 weeks, our model outperformed all S2-based studies and most of the other studies. A direct comparison is challenging, however, as the term "early detection" is often defined as green attack detection (infested trees that are still green), ignoring the exact time after infestation (probably due to the lack of reliable reference datasets).

The green attack, however, can last from a few weeks (Huo et al. 2022) up to a few months after the infestation (Bárta et al. 2021, Huo et al. 2021), depending on environmental conditions, for instance. Independent of that, the bark beetle brood swarms out ~6 to 10 weeks after the infestation (Kautz et al. 2024). Accordingly, forest managers are primarily interested in an early detection in terms of a maximum of 10 weeks after the infestation, regardless of whether the tree is still green or already red. Two studies are more explicit with respect to the temporal dimension. First, Mandl and Lang (2023) reported the detection date and performance relative to ground observations on 4-weekly surveys determining the infested trees as "freshly infested and still green". Most of the detected infestations in this study were between 3 weeks before or after the ground observation. While

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

this might imply an early detection among the correctly classified infestations, the study used *post hoc* information from the end of the study period for its detections (see below). This impairs direct applicability of the approach as a monitoring system. Second, Ortiz et al. (2013) used data from a pheromone attraction experiment set up at a specific date, and used single RapidEye (optical satellite) and TerraSAR-X (Radar) scenes acquired ~7 weeks after starting the experiment. Thus, these results imply an early detection, as long as there were no infestations before starting the experiment. However, besides the higher spatial resolution of the sensors used in this study, the training and test site was also limited to a single small forest stand in this study. This further facilitates successful classification, as the model did not have to deal with additional variability introduced by differing site conditions and forest structures.

The problem of limited extent of study areas applies to many earlier studies (e.g. Ortiz et al. 2013, Minařík et al. 2021, Dalponte et al. 2022, and Mandl and Lang 2023). Studies focusing on larger regions such as the Bavarian Forest National Park in Germany also did not test the spatial transferability of the model (Abdullah et al. 2019). Working with small study areas likely decreases the variability of both undisturbed and infested time series and image features (sun angle, etc.), which in turn provides much better conditions for separation of the classes and thus probably leads to an increase of model performance (cp. Kattenborn et al. 2022) as compared to an operational setting where large areas have to be monitored. To the best of our knowledge, Marx et al. (2024) is the only publication about early detection in the European spruce bark beetle versus Norway spruce system that used a training dataset far away from the test dataset (here: 436 km). The latter study achieved good results on early detection between 13 and 70 days after infestation using a hyperspectral sensor mounted on a UAV with spatial resolution of 5 cm. Therefore, this is the only study that suggests that early detections are feasible across large areas with the same model. Besides that, our study appears to be the only one which uses a spatial hold-out as test data, which provides a more pessimistic assessment, since environmental conditions (e.g. temperature, understory composition) and sensor effects (sun angle, etc.) may both differ.

Another drawback of former early detection studies is the use of information from the end of the detection period to distinguish between undisturbed and infested trees during the detection period for model or data calibration. For instance, Mandl and Lang (2023) retrospectively analyzed S2 scenes from 2020, using gray attack areas determined in September 2020 in aerial imagery to derive vegetation index (VI) thresholds and show the *post hoc* separability of green attack and undisturbed trees in the same year. Dalponte et al. (2022) used VI values from the last S2 scene of their study period as a baseline to calculate temporal changes in VI values, which were then used for classification. While these approaches demonstrate the potential of S2 for early or green attack detection, they are not ready for NRT monitoring. As opposed to that, our approach used automated thresholding on data from different areas and partly different years to infer a suitable decision threshold. This threshold demonstrated robustness, as anomalies clearly stand out against non-anomalous sequences in the time series (cp. Fig. 4 and Fig. 7).

Accordingly, to the best of our knowledge, this is the first study that applied an early bark beetle infestation detection approach readily applicable for NRT monitoring on free-of-charge S2 time series with spatially clearly distinct training and test areas and temporally very accurate infestation data. An operational remote sensing–based system successfully confining bark beetle outbreaks was estimated to require the timely removal of at least 80% of the infested trees (Pietzsch et al. 2023) while minimizing the labor needed to achieve this. Consequently, in light of our results and those from other studies discussed herein, this goal seems to be almost unachievable at the moment. Laborious terrestrial observations, as opposed to that, have shown UA of 91% and PA of 93% (Kautz et al. 2023), clearly meeting the accuracy requirement.

One reason that hinders early detection in an NRT large-area approach is likely the noise in the S2 time series. Despite a strict quality assurance (see section Methods), single timestep outliers may occur as point anomalies (cp. Fig. 7). Thus, we needed to allow for two consecutive anomalous observations to trigger a warning, which delays the detection time by at least one revisit cycle (two consecutive clear sky observations). In most cases, this leads to a delay of at least 5 days. Note, however, that the S2A and S2B orbits partly overlap, which leads to improved revisit times in the overlapping areas. Thus, the probability of clear sky observations is higher in these cases. Since the model has been trained on data covering large areas, the training dataset contains both overlapping and non-overlapping parts of the orbits. Therefore, the model should be able to cope with both enhanced and reduced sampling frequencies.

Apart from this general challenge of optical satellites in NRT monitoring, the spatial resolution might be one main drawback of S2 in early bark beetle detections. The best results in literature were achieved using satellites or UAV allowing for analysis on individual tree crown level, e.g. Ortiz et al. (2013), Immitzer and Atzberger (2014), and Minařík et al. (2021). Individual tree analysis ensures that the background signal, e.g. coming from soil, other vegetation, or undisturbed trees, does not overlap the spectral signal of the trees. Accordingly, the findings of Huo et al. (2022) suggest that 5 to 10 disturbed trees in a 15-m plot of ~30–50 trees are necessary to show any spectral response in S2. For most plots in this study, significant changes in S2 bands occurred only after ~3 months and only in severely infested plots (>10 disturbed trees).

While our anomaly detection approach bears great potential in early detection tasks, it requires a manual selection of the most suitable S2 bands or combinations thereof. One of the main advantages of most DL models over other modeling approaches, including other Machine Learning algorithms, is that manual feature engineering is not necessary. These DL models inherently engineer relevant features automatically from raw data and extract relevant representations of those features, often followed by a classification or regression (e.g. Yuan and Lin 2021, Schiefer et al. 2023, and Schiller et al. 2024). In our case, however, the approach was to reconstruct the input, meaning that automatic feature engineering can only take place internally, whereas the output was a reproduction of the input features. Thus, we needed to provide an input that is optimally suited for the bark beetle detection task. While many VIs have been developed and tested for this task, there is no agreement in literature about a specific index. Promising VIs are designed either to detect changes in cell structure and chlorophyll content (Abdullah et al. 2019, Marx et al. 2024) or leaf water content (Abdullah et al. 2019, Mandl and Lang 2023). The raw SWIR1 band can also be used for forest disturbance detection (Grabska et al. 2020) and has been shown to be sensitive to bark beetle infestations in many studies (Abdullah et al. 2019, Huo et al. 2021, Huo et al. 2022). Since preliminary tests indicated best performance of CR_{SWIR}, we used it here. The CR_{SWIR} index, which has not been used frequently in literature yet, amplifies subtle changes in vegetation water content compared to SWIR1

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

alone. This has shown to be optimal for this disturbance detection task. For instance, FORDEAD (Dutrieux et al. 2021), an NRT disturbance detection approach using S2 time series, detected 68% of green attacks on spruce stands (Dutrieux et al. 2021). The CR_{SWIR} index has also been used to achieve a drought disturbance detection in oak trees using S2 time series (Mouret et al. 2024).

The mentioned FORDEAD approach fits a harmonic model to the S2 time series of the pixel of interest during a training period and compares observations to model predictions in the consecutive detection period. A potential advantage of FORDEAD might be that the model only has to account for the variability within a single pixel, whereas our model has to internalize the spectral variability of all of the training time series across Central Europe. Consequently, our model might be less sensitive to subtle changes in the spectral trajectory which might constitute normal spectral patterns considering large areas, but are a sign of bark beetle infestation for a time series at a specific location.

Considering that local/regional studies tend to achieve better results (see above), it might be necessary to add local information to the model to reduce the variability of what constitutes an undisturbed time series. In the current model, regional differences, e.g. in the phenology, were largely ignored. One option to improve upon this is to use thermal temporal encoding (Nyborg et al. 2022), e.g. the cumulative soil temperature across the year as observed by and extrapolated from ground-based local weather station records, instead of our day-of-the-year encoding. Meteorological conditions strongly influence the phenology of trees, e.g. by shortening the growing season at higher altitudes. Using extrapolated or modeled temperature information from weather station records, the variability of the features of an undisturbed time series could be constrained. Such local cues might support an earlier detection of bark beetle infestations.

A further option would be to couple the results of the presented model with a predisposition approach. There is evidence that the susceptibility of spruce trees to bark beetle infestations can be modeled using various environmental factors. These include stand characteristics such as age and density, site conditions like drought stress, and climate variables such as temperature and precipitation (Overbeck and Schmidt 2012, Netherer et al. 2019, Xu et al. 2024). For the determined predisposed stands, a lower anomaly threshold could be used to achieve an earlier detection, or they could be prioritized in case of limited resources for ground observations.

Conclusion

In this study, we used DL-based reconstructions of S2 time series to detect anomalies in the CR_{SWIR} index as an attempt for early detections of bark beetle infestations in Central Europe. For model training, the approach requires only undisturbed forest as reference data, which are generally easier to acquire, as most of the forest in Central Europe remains intact despite numerous disturbances. While a satisfactory amount of early detections appears impossible with our approach, the model seems well suited for forest disturbance detection in general. We provide trained models and code at https://github.com/ChrSchiller/TimesNet_RSAD also including all other forest types (broadleaved, mixed, coniferous) and the 10 S2 bands with 10- and 20-m spatial resolution for the scientific community to apply in following studies. All trained models, however, should be evaluated on a broader scale and scope (e.g. different disturbance agents) to verify a comprehensive applicability since the current test dataset was confined to a small area and a short time period of disturbance events.

Upcoming improved NRT monitoring approaches might need to use regional models or additional regional information to reduce the spectral variability of non-infested trees for a better separability of subtle changes in the spectral trajectory. Additionally, the model performance could be enhanced by additional information from a predisposition approach that models the susceptibility of the spruce trees in each pixel based upon environmental variables. Moreover, sensors with higher spatial resolution might be used, as they can be applied at individual tree crown level, generally achieving better results (Ortiz et al. 2013, Immitzer and Atzberger 2014, Minařík et al. 2021). Data from these sensors, however, are usually cost-intensive and thus not yet readily available for an area-wide monitoring across large areas.

All in all, our approach did not achieve a sufficient accuracy to replace terrestrial bark beetle surveys for early detection, but appears very promising when given more time for the detection (e.g. 13 weeks). As we provide the trained model for a ready-to-use NRT bark beetle monitoring system, it may support ground observations by giving hints at specific locations that might have been missed by human observers. Consequently, it may complement terrestrial surveys in future operational use.

Acknowledgements

We thank Selina Schwarz from Karlsruhe Institute of Technology (KIT) for providing the forest dieback dataset of Luxembourg and Berthold Mertens from Wald und Holz NRW for approving the use of the test dataset. We are generally thankful for any efforts to publish datasets that can be used by other researchers, e.g. the FNEWS dataset from Thünen Institute in Eberswalde, Germany. We thank any institution that offers publicly available data, such as some of the German federal states and the state of Luxembourg. We greatly appreciate the provision of the S2 datacube of Germany preprocessed with the FORCE software on the platform www.eo-lab.org. Next to the data, code of the model architectures are needed to try newly developed approaches. Hence, we thank anyone who publishes the code along with their research. In case of this study, we dedicate our gratitude to the developers of the TimesNet model, which is the backbone of our approach. The editors and reviewers who commented on the manuscript greatly helped to improve it—thank you for that.

Author contributions

Christopher Schiller (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing—original draft), Johannes May (Data curation, Writing—review & editing), Randolf Klinke (Writing—review & editing), and Fabian Ewald Fassnacht (Funding acquisition, Project administration, Supervision, Writing—review & editing)

Supplementary data

Supplementary data are available at *Forestry* online.

Conflict of interest: None declared.

Funding

This work was supported by the German Federal Ministry for Environment, Nature Conservation, Nuclear Safety and Consumer Protection in the Future Forest project under grant number 67KI21002C.

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

Data availability

The test dataset can be made available upon request to the corresponding author, co-author J.M., or at the following data repository: https://zenodo.org/records/10495412. Most of the training data are available either openly (FNEWS: https://www.openagrar.de/receive/openagrar_mods_00093065) or upon request to the corresponding author (*own work* datasets), e.g. via the following data repository: https://zenodo.org/records/8397141. The Sentinel-2 datacubes of Austria and Luxembourg can be made available upon request.

References

Abdullah H, Darvishzadeh R, Skidmore AK. *et al.* European spruce bark beetle (Ips typographus, L.) green attack affects foliar reflectance and biochemical properties. *Int J Appl Earth Observ Geoinform* 2018;**64**:199–209. https://doi.org/10.1016/j.jag.2017.09.009.

Abdullah H, Skidmore AK, Darvishzadeh R. *et al.* Sentinel-2 accurately maps green-attack stage of European spruce bark beetle (*Ips typographus*, L.) compared with Landsat-8. *Remote Sens Ecol Conserv* 2019;**5**:87–106. https://doi.org/10.1002/rse2.93.

Bárta V, Lukeš P, Homolová L. Early detection of bark beetle infestation in Norway spruce forests of Central Europe using Sentinel-2. *Int J Appl Earth Observ Geoinform* 2021;**100**:102335. https://doi.org/10.1016/j.jag.2021.102335.

Bigler C, Gavin DG, Gunning C. *et al.* Drought induces lagged tree mortality in a subalpine Forest in the Rocky Mountains. *Oikos* 2007;**116**:1983–94. https://doi.org/10.1111/j.2007.0030-1299.16034.x.

Caudullo G, Tinner W, de Rigo D. Picea abies in Europe: distribution, habitat, usage and threats. In: San-Miguel-Ayanz J, de Rigo D, Caudullo G, Houston Durrant T, Mauri A (eds), *European Atlas of Forest Tree Species*. Luxembourg: Publications Office of the European Union, 2016.

Dalponte M, Solano-Correa YT, Frizzera L. *et al.* Mapping a European spruce bark beetle outbreak using Sentinel-2 remote sensing data. *Remote Sens* 2022;**14**:3135. https://doi.org/10.3390/rs14133135.

Du B, Yuan Z, Bo Y. *et al.* A combined deep learning and prior knowledge constraint approach for large-scale forest disturbance detection using time series remote sensing data. *Remote Sens* 2023;**15**:2963. https://doi.org/10.3390/rs15122963.

Dutrieux R, Féret J-B, Ose K. Mise au point d'une méthode reproductible pour le suivi généralisé des dégats de scolytes par télédétection satellitaire. *ONF Rend Techn* 2021;**69-70**:37–44.

Dutrieux R, Féret J-B, Ose K. *et al.* Package Fordead. 2021.

Federal Ministry of Food and Agriculture (BMEL). *Der Wald in Deutschland - Ergebnisse der Vierten Bundeswaldinventur (the Forest in Germany - Results of the Fourth National Forest Inventory)*. Bonn, Germany: Federal Ministry of Food and Agriculture, 2024.

Frantz D. FORCE—Landsat + Sentinel-2 analysis ready data and beyond. *Remote Sens* 2019;**11**:1124. https://doi.org/10.3390/rs11091124.

Frantz D, Haß E, Uhl A. *et al.* Improvement of the Fmask algorithm for Sentinel-2 images: separating clouds from bright surfaces based on parallax effects. *Remote Sens Environ* 2018;**215**:471–81. https://doi.org/10.1016/j.rse.2018.04.046.

García-Herrera R, Garrido-Perez JM, Barriopedro D. *et al.* The European 2016/17 drought. *J Climate* 2019;**32**:3169–87. https://doi.org/10.1175/JCLI-D-18-0331.1.

Grabska E, Hawryło P, Socha J. Continuous detection of small-scale changes in Scots pine dominated stands using dense Sentinel-2 time series. *Remote Sens* 2020;**12**:1298. https://doi.org/10.3390/rs12081298.

Grillakis MG. Increase in severe and extreme soil moisture droughts for Europe under climate change. *Sci Total Environ* 2019;**660**:1245–55. https://doi.org/10.1016/j.scitotenv.2019.01.001.

Haberstroh S, Werner C, Grün M. *et al.* Central European 2018 hot drought shifts Scots pine forest to its tipping point. *Plant Biol J* 2022;**24**:1186–97. https://doi.org/10.1111/plb.13455.

Hijmans RJ. terra: Spatial Data Analysis. 2023.

Huo L, Lindberg E, Fransson JES. *et al.* Comparing spectral differences between healthy and early infested spruce forests caused by bark beetle attacks using satellite images. In: *IGARSS 2022–2022 IEEE International Geoscience and Remote Sensing Symposium*, edited by IEEE. Kuala Lumpur, Malaysia: IEEE, 2022, 7709–12.

Huo L, Persson HJ, Lindberg E. Early detection of forest stress from European spruce bark beetle attack, and a new vegetation index: normalized distance red & SWIR (NDRS). *Remote Sens Environ* 2021;**255**:112240. https://doi.org/10.1016/j.rse.2020.112240.

Immitzer M, Atzberger C. Early detection of bark beetle infestation in Norway spruce (Picea abies, L.) using WorldView-2 data. *PFG* 2014;**2014**:351–67. https://doi.org/10.1127/1432-8364/2014/0229.

Ionita M, Nagavciuc V. Changes in drought features at the European level over the last 120 years. *Nat Hazards Earth Syst Sci* 2021;**21**:1685–701. https://doi.org/10.5194/nhess-21-1685-2021.

Kattenborn T, Schiefer F, Frey J. *et al.* Spatially autocorrelated training and validation samples inflate performance assessment of convolutional neural networks. *ISPRS Open J Photogramm Remote Sens* 2022;**5**:100018. https://doi.org/10.1016/j.ophoto.2022.100018.

Kautz M. On correcting the time-lag bias in aerial-surveyed bark beetle infestation data. *For Ecol Manage* 2014;**326**:157–62. https://doi.org/10.1016/j.foreco.2014.04.010.

Kautz M, Feurer J, Adler P. Early detection of bark beetle (Ips typographus) infestations by remote sensing – a critical review of recent research. *For Ecol Manage* 2024;**556**:121595. https://doi.org/10.1016/j.foreco.2023.121595.

Kautz M, Peter FJ, Harms L. *et al.* Patterns, drivers and detectability of infestation symptoms following attacks by the European spruce bark beetle. *J Pest Sci* 2023;**96**:403–14. https://doi.org/10.1007/s10340-022-01490-8.

Kirisits T, Offenthaler I. Xylem sap flow of Norway spruce after inoculation with the blue-stain fungus Ceratocystis polonica. *Plant Pathol* 2002;**51**:359–64. https://doi.org/10.1046/j.1365-3059.2002.00722.x.

Kokaly RF, Asner GP, Ollinger SV. *et al.* Characterizing canopy biochemistry from imaging spectroscopy and its application to ecosystem studies. *Remote Sens Environ* 2009;**113**:S78–91. https://doi.org/10.1016/j.rse.2008.10.018.

König S, Thonfeld F, Förster M. *et al.* Assessing combinations of Landsat, Sentinel-2 and Sentinel-1 time series for detecting bark beetle infestations. *GIScience Remote Sens* 2023;**60**:2226515. https://doi.org/10.1080/15481603.2023.2226515.

Kosma C, Nikolentzos G, Vazirgiannis M. Time-parameterized convolutional neural networks for irregularly sampled time series. Preprint on arXiv. 2023. https://doi.org/10.48550/arXiv.2308.03210.

Künniger T, Elsener R, Heeb M. *et al.* Mechanical properties of Norway spruce (Picea abies) infested by the bark beetle (Ips typographus). *Wood Mater Sci Eng* 2024;**19**:85–91. https://doi.org/10.1080/17480272.2023.2226644.

Langner N, Puhm M, Deutscher J. *et al. FNEWS-Jahresprodukte 2018 bis 2022*. Eberswalde, Germany: Thünen-Institut, 2023. Available from OpenAgrar.

Mandl L, Lang S. Uncovering early traces of bark beetle induced forest stress via semantically enriched Sentinel-2 data and

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

spectral indices. *PFG* 2023;**91**:211–31. https://doi.org/10.1007/s41064-023-00240-4.

Marvasti-Zadeh SM, Goodsman D, Ray N. *et al.* Early detection of bark beetle attack using remote sensing and machine learning: a review. *ACM Computing Surveys* 2024;**56**:Article 97.

Marx A, Clasen A, May J. *et al.* Imaging spectroscopy for bark beetle detection in Norway spruce and the relevance of the red-edge spectral range. *Int J Appl Earth Observ Geoinform* 2024;**133**:104100. https://doi.org/10.1016/j.jag.2024.104100.

Minařík R, Langhammer J, Lendzioch T. Detection of bark beetle disturbance at tree level using UAS multispectral imagery and deep learning. *Remote Sens* 2021;**13**:4768. https://doi.org/10.3390/rs13234768.

Mouret F, Morin D, Martin H. *et al.* Toward an operational monitoring of oak dieback with multispectral satellite time series: a case study in Centre-val De Loire region of France. *IEEE J Sel Top Appl Earth Observ Remote Sens* 2024;**17**:643–59. https://doi.org/10.1109/JSTARS.2023.3332420.

Netherer S, Panassiti B, Pennerstorfer J. *et al.* Acute drought is an important driver of bark beetle infestation in Austrian Norway spruce stands. *Front For Glob Change* 2019;**2**:39. https://doi.org/10.3389/ffgc.2019.00039.

Nyborg J, Pelletier C, Assent I. Generalized classification of satellite image time series with thermal positional encoding. In: *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, edited by IEEE. New Orleans, LA, USA: IEEE, 2022, 1391–401.

Ortiz S, Breidenbach J, Kändler G. Early detection of bark beetle green attack using TerraSAR-X and RapidEye data. *Remote Sens* 2013;**5**:1912–31. https://doi.org/10.3390/rs5041912.

Overbeck M, Schmidt M. Modelling infestation risk of Norway spruce by Ips typographus (L.) in the Lower Saxon Harz Mountains (Germany). *For Ecol Manage* 2012;**266**:115–25. https://doi.org/10.1016/j.foreco.2011.11.011.

Paszke A, Gross S, Massa F. *et al.* PyTorch: an imperative style. In: *Advances in Neural Information Processing Systems*. Vancouver, Canada: High-Performance Deep Learning Library; 2019;**32**:8024–35.

Patacca M, Lindner M, Lucas-Borja ME. *et al.* Significant increase in natural disturbance impacts on European forests since 1950. *Glob Chang Biol* 2022;**29**:1359–76. https://doi.org/10.1111/gcb.16531.

Pebesma E. Simple features for R: standardized support for spatial vector data. *R J* 2018;**10**:439–46. https://doi.org/10.32614/RJ-2018-009.

Perbet P, Guindon L, Côté J-F. *et al.* Evaluating deep learning methods applied to Landsat time series subsequences to detect and classify boreal forest disturbances events: the challenge of partial and progressive disturbances. *Remote Sens Environ* 2024;**306**:114107. https://doi.org/10.1016/j.rse.2024.114107.

Pietzsch BW, Wudel C, Berger U. Nonparametric upscaling of bark beetle infestations and management from plot to landscape level by combining individual-based with Markov chain models. *Eur J Forest Res* 2023;**142**:129–44. https://doi.org/10.1007/s10342-022-01512-1.

Puhm M, Deutscher J, Hirschmugl M. *et al.* A near real-time method for forest change detection based on a structural time series model and the Kalman filter. *Remote Sens* 2020;**12**:3135. https://doi.org/10.3390/rs12193135.

R Core Team. R: A Language and Environment for Statistical Computing. 2023.

Reinosch E, Backa J, Adler P. *et al.* Detailed validation of large-scale Sentinel-2-based forest disturbance maps across Germany. *Forestry* 2025;**98**:437–453.

Schiefer F, Kattenborn T, Frick A. *et al.* Mapping forest tree species in high resolution UAV-based RGB-imagery by means of convolutional neural networks. *ISPRS J Photogramm Remote Sens* 2020;**170**:205–15. https://doi.org/10.1016/j.isprsjprs.2020.10.015.

Schiefer F, Schmidtlein S, Frick A. *et al.* UAV-based reference data for the prediction of fractional cover of standing deadwood from Sentinel time series. *ISPRS Open J Photogramm Remote Sens* 2023;**8**:100034. https://doi.org/10.1016/j.ophoto.2023.100034.

Schiefer F, Schmidtlein S, Hartmann H. *et al.* Large-scale remote sensing reveals that tree mortality in Germany appears to be greater than previously expected. *Forestry* 2024;**98**:535–549.

Schiller C, Költzow J, Schwarz S. *et al.* Forest disturbance detection in Central Europe using transformers and Sentinel-2 time series. *Remote Sens Environ* 2024;**315**:114475. https://doi.org/10.1016/j.rse.2024.114475.

Schwarz S, Werner C, Fassnacht FE. *et al.* Forest canopy mortality during the 2018-2020 summer drought years in Central Europe: the application of a deep learning approach on aerial images across Luxembourg. *Forestry* 2024;**97**:376–87. https://doi.org/10.1093/forestry/cpad049.

Senf C, Seidl R, Hostert P. Remote sensing of forest insect disturbances: current state and future directions. *Int J Appl Earth Observ Geoinform* 2017;**60**:49–60. https://doi.org/10.1016/j.jag.2017.04.004.

Solheim H. Fungal succession in sapwood of Norway spruce infested by the bark beetle Ips typographus. *Eur J Forest Pathol* 1992;**22**:136–48. https://doi.org/10.1111/j.1439-0329.1992.tb01440.x.

Ustin SL, Roberts DA, Gamon JA. *et al.* Using imaging spectroscopy to study ecosystem processes and properties. *BioScience* 2004;**54**:523. https://doi.org/10.1641/0006-3568(2004)054[0523:UISTSE]2.0.CO;2.

Van Rossum G, Drake FL. *Python 3 Reference Manual*. Beaverton, OR: Python Software Foundation, 2009.

Verbesselt J, Zeileis A, Herold M. Near real-time disturbance detection using satellite image time series. *Remote Sens Environ* 2012;**123**:98–108. https://doi.org/10.1016/j.rse.2012.02.022.

Vošvrdová N, Johansson A, Turčáni M. *et al.* Dogs trained to recognise a bark beetle pheromone locate recently attacked spruces better than human experts. *For Ecol Manage* 2023;**528**:120626. https://doi.org/10.1016/j.foreco.2022.120626.

Wu H, Hu T, Liu Y *et al. TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis*. Preprint on arXiv, 2023. https://doi.org/10.48550/arXiv.2210.02186.

Xu C, Förster M, Gränzig T. *et al.* Relating soil moisture and Sentinel-2 vegetation index patterns to spruce bark beetle infestations prior to outbreak. *Forestry* 2024;**97**:728–738.

Ye S, Rogan J, Zhu Z. *et al.* A near-real-time approach for monitoring forest disturbance using Landsat time series: stochastic continuous change detection. *Remote Sens Environ* 2021;**252**:112167. https://doi.org/10.1016/j.rse.2020.112167.

Yuan Y, Lin L. Self-supervised pretraining of transformers for satellite image time series classification. *IEEE J Sel Top Appl Earth Observ Remote Sens* 2021;**14**:474–87. https://doi.org/10.1109/JSTARS.2020.3036602.

Zhu Z, Woodcock CE. Continuous change detection and classification of land cover using all available Landsat data. *Remote Sens Environ* 2014;**144**:152–71. https://doi.org/10.1016/j.rse.2014.01.011.

Downloaded from academic.oup.com/forestry/article/99/2/cpaf053/8250836 by guest on 18 October 2026

© The Author(s) 2025. Published by Oxford University Press on behalf of the Institute of Chartered Foresters. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.
Forestry: An International Journal of Forest Research, 2026, **99**, 1–14
https://doi.org/10.1093/forestry/cpaf053
Original Article